

EXERCISE 2: 'SQL FUNDAMENTALS'

1. SELECT Statement

---> Write a SQL query to retrieve a columns from the employee table.

SELECT *

FROM employee-table,

Id	first-name	last-name	department	Salary	hire-date	City
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johanson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-08-25	New York
5	David	White	Marketing	52000	2016-04-10	San Francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Garcia	Marketing	58000	2022-06-01	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

2. SELECT DISTINCT Statement

2.1 Write a SQL query to find all the unique department in the employee table.

→ SELECT department,
FROM employee-table,

department
IT
HR
FINANCE
MARKETING

3. ORDER BY statement

3.1. Write a SQL query to retrieve all employees' first & last names, ordered by salary in descending order.

→ SELECT first-name, last-name, salary
FROM employees
ORDER BY salary DESC

First-name	last-name	Salary
Emily	Davis	62000
Mike	Johnson	60000
Robert	Wilson	59000
John	Doe	55000
Daniel	Clark	53000
Sarah	Brown	53000
David	White	52000
Jessica	Moore	51000
Laura	Hall	50000
Jane	Smith	48000

4. LIMIT Statement

4.1. Write a SQL query to retrieve the 5 highest-paid employees

→ SELECT ~~employees~~,
first-name,
last-name,
salary

FROM employee
ORDER BY salary DESC
LIMIT 5;

first-name	last-name	Salary
Emily	Davis	62000
Mike	Johnson	60000
Robert	Wilson	59000
John	Doe	55000
Daniel	Clark	53000

5. WHERE Statement

5.1. Write a SQL query to find employees who work in the IT department.

→ **SELECT** employee-id, first-name, last-name, department
FROM employee
WHERE department = 'IT';

Id	first-name	last-name	department
1	John	Dee	IT
4	Sarah	Brown	IT
6	Emily	Davis	IT
10	Laura	Hall	IT

6. AND Statement

6.1. Write a SQL query to find employee who work in the ^{Finance} department AND have a salary greater than 58,000.

→ **SELECT** employee-id,
first-name,
last-name,
department,
Salary.

FROM employee

WHERE department = 'Finance' AND salary > 58000;

Id	first-name	last-name	department	Salary
3	Mike	Johnson	Finance	60000
7	Robert	Wilson	Finance	59000

7. OR Statement

7.1. Write a SQL query to find employees who ^{work} ~~do not~~ in the HR department OR the Marketing department.

→ **SELECT** employee-id,
first-name,
last-name,
department

FROM employees

WHERE department = 'HR' OR 'Marketing';

OR Statement

Id	first-name	last-name	department
2	Jane	Smith	HR
5	David	White	Marketing
8	Jessica	Moore	HR
9	Daniel	Clark	Marketing

8. NOT Statement

8.1. Write a SQL query to find employees who do not work in the IT department.

→ `SELECT` ^{employee_id} first-name
^{last-name}
^{department}

`FROM` employees

`WHERE` department `NOT IN` ('IT');

Id	first-name	last-name	department
2	Jane	Smith	HR
3	Mike	Johanson	Finance
5	David	White	Marketing
7	Robert	Wilson	Finance
8	Jessica	Moore	HR
9	Daniel	Clark	Marketing

9. IN Statement

9.1. Write a SQL query to find employees who are IN the HR, IT, or Finance department.

— `SELECT` ^{employee_id}
^{first-name}
^{last-name}
^{department}

`FROM` employees

`WHERE` department `IN` ('HR', 'IT', 'FINANCE');

IN Statement

Id	first name	last name	department
2	Jane	Smith	HR
8	Jessica	Moore	HR
1	John	DOE	IT
4	Sarah	Brown	IT
6	Emily	Davis	IT
10	Laura	Hall	IT
3	Mike	Johnson	FINANCE
7	Robert	Wilson	FINANCE

10. COMBINING CONDITIONS

10.1. Write SQL query to find employees who are in the IT department, have a salary greater than 50000 AND are located in New York.

→ SELECT employee_id,
first_name,
department, last_name,
salary,
city

FROM employees

WHERE department = 'IT'

AND salary > 50000

AND city = 'New York';

Id	first name	last name	department	Salary	City
1	John	DOE	IT	55000	New York
4	Sarah	Brown	IT	53000	New York.

11.

11. Combining WHERE, AND, and ORDER BY

11.1 Write a SQL query to retrieve the first and last names of employees who work in the Finance or Marketing department earn more than 52000, and ORDER the results by salary in descending order.

→ SELECT employee_id,
first_name,
last_name,
department,
salary

FROM employees

WHERE department IN ('Finance', 'Marketing')

AND salary > 52000

ORDER BY salary DESC;

ID	first_name	last_name	department	Salary
3	Mike	Johnson	Finance	60000
5	David	White	Marketing	52000
7	Robert	Wilson	Finance	59000
9	Daniel	Clark	Marketing	53000

12. Combining WHERE, NOT, AND

12. Combining SELECT DISTINCT, WHERE and IN

12.1 Write a SQL query to find all the unique cities where employees work, excluding those in the IT and HR department.

→ SELECT DISTINCT city

FROM employees

WHERE department NOT IN ('IT', 'HR');

city	department
Los Angeles	Finance
San Francisco	Marketing
Houston	Finance

13. COMBINING WHERE, NOT, AND, and ORDER BY

13.1 Write a SQL query to retrieve employees who are NOT in the FINANCE department, have a salary greater than 50000 and order the results by hire date in ascending order.

→ SELECT *

FROM employee

WHERE department <> 'FINANCE'

AND salary > 50000

ORDER BY hire-date ASC;

Id	first-name	last-name	department	salary	hire-date	City
6	Emily	Davis	IT	62000	2015-02-14	Chicago
5	David	White	Marketing	52000	2016-04-10	San Francisco
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
1	John	Boe	IT	55000	2018-06-15	New York
4	Sarah	Brown	IT	53000	2021-03-25	New York
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago

14. Write a SQL query to find the first 3 employees who work in either CHICAGO OR LOS ANGELES and belong to IT or MARKETING department.

→ SELECT *

FROM employees

WHERE city IN ('CHICAGO', 'Los Angeles')

AND department IN ('IT', 'Marketing')

LIMIT 3;

Id	first-name	last-name	department	salary	hire-date	City
1	John	Boe	IT	55000	2018-06-15	N
6	Emily	Davis	IT	62000	2015-02-14	CHICAGO
9	Daniel	Clark	Marketing	53000	2022-06-01	CHICAGO